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Calculating the CMB power spectrum from simulated cosmic structure formation initiated by primordial fluctuations

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ABSTRACT

I will write an abstract for the whole project at the end, once I have completed all the milestones.

Key words. cosmic microwave background – large-scale structure of Universe – linear perturbations

0. Introduction

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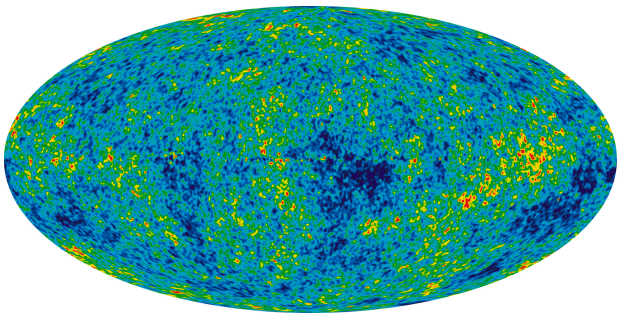

Fig. 1. CMB (NASA WMAP Science Team 2025)

1. Background cosmology

To compute the CMB power spectrum, we must first establish the background cosmological model upon which perturbations will evolve. This section details the construction of this background Universe. We begin by outlining the theoretical framework, most notably the Friedmann-Lemaître-Robertson-Walker (FLRW) metric and the resulting Friedmann equations that governs the expansion history and the evolution of its energy components. Subsequently, we describe the numerical implementation of this model, including the fiducial parameters adopted from Planck 2018. Finally, we validate our implementation by comparing its predicted timeline of cosmic epochs and derived quantities, such as the age of the universe, against theoretical expectations, and we present the results of an MCMC parameter fit to supernova data, which contextualizes the well-known Hubble tension within our model.

* The code developped for this project is available at GitHub.

1.1. Theory

In this section, we will present the relevant theory for developping the background cosmology in our model Universe along with all of the equations used. All of the theory is covered in greater detail in the lecture notes by Hans A. Winther for the cosmology course AST5220 (Winther 2026a,b), so we will simply summarise the key details and derive some quantities that will come in handy for the numerical implementation.

1.1.1. The Friedmann-Lemaître-Robertson-Walker geometry

For our background cosmology, we assume an isotropic and homogeneous Universe with a spatial scale factor $a(t)$ equipped with the Friedmann-Lemaître-Robertson-Walker (FLRW) metric

$$ds^2 = -dt^2 + a^2(t)\Big(dx^2 + dy^2 + dz^2\Big), \tag{1}$$

or, on component form; $g_{00} = -1$, $g_{ij} = a^2(t)\delta_{ij}$.

At the background level, the constituents of our Universe, which include radiation (photons and massless neutrinos), non-relativistic matter (baryons and cold dark matter), and a cosmological constant, are treated as perfect fluids. For each component i, the stress-energy tensor in the comoving frame takes the diagonal form

$$T^{(i)}_{\mu\nu} = \mathrm{diag}(\rho_i, p_i, p_i, p_i),$$

with ρ_i denoting the density and p_i the pressure of that component.

The resulting Einstein field equations for the FLRW metric and the sum of the stress-energy tensors for the perfect fluids,

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G \sum_i T^{(i)}_{\mu\nu},$$

we get the Friedmann equations. The (00) component of the Einstein equations gives us the first equation,

$$H^2(a) \equiv \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\sum_i \rho_i, \tag{2}$$

while the (11), (22) and (33) components gives us the second Friedmann equation (assuming isotropy, or else they would yield three different equations),

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\sum_i(\rho_i + 3p_i). \tag{3}$$

As our time variable, we will use the logarithm of scale factor $x \equiv \ln[a(t)]$. We thus need to study the Friedmann equations, Eqs.(2)-(3), to understand how the scale factor evolves and how it is related to the evolution of the densities of the different components in our model Universe.

1.1.2. The Hubble function

Introducing the definition of the density parameters $\Omega_i \equiv 8\pi G\rho_i/3H^2$, and the equation of state $w_i \equiv p_i/\rho_i$, the first Friedmann equation can be recast into the form we will be using for our implementation of the background cosmology

$$H^2 = H_0^2\sum_i \frac{\Omega_{i0}}{a^{3(1+w_i)}},$$

where $H_0 = 100h$ km/s/Mpc is the Hubble constant, and Ω_{i0} is the current day value of the density parameter of component i.

In our model Universe, we have dark energy, described by a cosmological constant (Λ) with equation of state $w_\Lambda = -1$, non-relativistic matter in the form of baryons (b) and cold dark matter (CDM) with equation of state $w_M = 0$, and two relativistic species, photons (γ) and neutrinos (ν) with $w_R = 1/3$. The curvature (k) can be decribed as having an equation of state $w_k = -1/3$. We can thus explicitly write out our Hubble function as

$$H(a) = H_0\sqrt{\Omega_{\Lambda 0} + \frac{\Omega_{k0}}{a^2} + \frac{\Omega_{b0} + \Omega_{\rm CDM0}}{a^3} + \frac{\Omega_{\gamma 0} + \Omega_{\nu 0}}{a^4}}. \tag{4}$$

The evolution of the density parameters as functions of the scale factor are thus given by

$$\Omega_k(a) = \frac{\Omega_{K0}}{a^2H(a)^2/H_0^2}, \qquad \Omega_M(a) = \frac{\Omega_{M0}}{a^3H(a)^2/H_0^2},$$
$$\Omega_R(a) = \frac{\Omega_{R0}}{a^4H(a)^2/H_0^2}, \qquad \Omega_\Lambda(a) = \frac{\Omega_{\Lambda 0}}{H(a)^2/H_0^2}. \tag{5}$$

Here we define the non-relativistic matter density parameter $\Omega_M = \Omega_b + \Omega_{\rm CDM}$ and the relativistic species density parameter $\Omega_R = \Omega_\gamma + \Omega_\nu$.

1.1.3. Cosmic and conformal time

In the FLRW metric, Eq.(1), we use the *cosmic time* coordinate t which is related to the scale factor through the Hubble function through the relation

$$H \equiv \frac{1}{a}\frac{\mathrm{d}a}{\mathrm{d}t} \;\Rightarrow\; \frac{\mathrm{d}t}{\mathrm{d}x} = \frac{1}{H},$$

or on integral form

$$t(x) = \int_{-\infty}^{x}\frac{\mathrm{d}x'}{H(x')}. \tag{6}$$

However, we also consider the *conformal time* η which is the distance that light may have travelled since the Big Bang (horizon). It is related to the cosmic time thorugh the relation

$$a\mathrm{d}\eta = c\mathrm{d}t \;\Rightarrow\; \frac{\mathrm{d}\eta}{\mathrm{d}x} = \frac{c}{aH}.$$

For calculating η, it is also prudent to introduce the conformal Hubble function $\mathcal{H} \equiv \frac{1}{a}\frac{\mathrm{d}a}{\mathrm{d}\eta} = aH$ such that the conformal time can be written on integral form as

$$\eta(x) = \int_{-\infty}^{x}\frac{c\mathrm{d}x'}{\mathcal{H}(x')}. \tag{7}$$

The derivatives of $\mathcal{H}$ with respect to x are given by;

$$\frac{\mathrm{d}\mathcal{H}}{\mathrm{d}x} = \mathcal{H}\left[1 - \frac{3}{2}\Omega_M - 2\Omega_R - \Omega_K\right]\Bigg|_{a=e^x}, \tag{8}$$

$$\frac{\mathrm{d}^2\mathcal{H}}{\mathrm{d}x^2} = \mathcal{H}\Bigg\{\left[1 - \frac{3}{2}\Omega_M - 2\Omega_R - \Omega_K\right]^2 - \frac{1}{2}\left(3\Omega_M + 4\Omega_R + 2\Omega_K\right)^2 + \frac{1}{2}\left(9\Omega_M + 16\Omega_R + 4\Omega_K\right)\Bigg\}\Bigg|_{a=e^x}, \tag{9}$$

where Ω_i are the density parameters evaluated at $a = e^x$. The derivatives of $\mathcal{H}$ are derived in Appendix A.

1.1.4. Luminosity distance and angular diameter distance

In order to compare our model with observations, and be able to perform parameter fitting, we will need to calculate two observables; namely the *angular diameter distance* d_A, and the *luminosity distance* d_L.

Writing the line element in the FLRW metric, Eq.(1), using spherical coordinates, we have

$$\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + a^2\left(\frac{\mathrm{d}r^2}{1-kr^2} + r^2\mathrm{d}\theta^2 + r^2\sin\theta^2\mathrm{d}\phi^2\right), \tag{10}$$

where k is the curvature. Since we are observing photons (that move radially along the line of sight) when determining astrophysical distances, we consider the null geodesic, $\mathrm{d}s^2 = 0$, followed by the rays of light. For these photons, we get

$$c\mathrm{d}t = \frac{a\mathrm{d}r}{\sqrt{1-kr}} \;\Rightarrow\; \chi \equiv \eta_0 - \eta = \int_0^r\frac{\mathrm{d}r'}{\sqrt{1-kr'}}. \tag{11}$$

Here, χ denotes the *comoving distance*.

Solving the integral in Eq.(11) and isolating r, we find three solutions, depending on the curvature

$$r = \begin{cases} \dfrac{\chi\sin\left[\varpi(\Omega_{K0})\right]}{\varpi(\Omega_{K0})} & \Omega_{k0} < 0, \\ \chi & \Omega_{k0} = 0, \\ \dfrac{\chi\sinh\left[\varpi(\Omega_{K0})\right]}{\varpi(\Omega_{K0})} & \Omega_{k0} > 0, \end{cases}$$

where $\varpi(\Omega_{K0}) \equiv \sqrt{|\Omega_{k0}|}H_0\chi/c$.

If an object has physical size D and subtends an angle θ on the sky, the angular diameter distance is defined as $d_A = D/\theta$. Reading off the line element, Eq.(10), a displacement in the θ-direction corresponds to a proper distance $dD = ar\,d\theta$, so

$$d_A = D/\theta = ar.$$

On the other hand, from the intrinsic luminosity of a source and its observed flux, the luminosity distance d_L can be defined through $F = L/4\pi d_L^2$, which simply reads

$$d_L = a/r = d_A/a^2.$$

1.1.5. Some cosmological events

Finally, for our analysis of the simulated data, it is convenient to identify key cosmological events to serve as heuristic markers for our interpretation of the results. We will look at three relevant events at the time scale of the evolution of the background:

(i) matter-radiation equality at $x_{\rm eq}^{MR} = \ln\left(\frac{\Omega_{R0}}{\Omega_{M0}}\right)$;

(ii) the onset of cosmic acceleration (when $\ddot{a} = 0$) at

$$x_{\rm acc} = \frac{1}{3}\ln\left(\frac{\Omega_{M0}}{2\Omega_{\Lambda 0}}\right);$$

(iii) and the matter-dark energy equality at

$$x_{\rm eq}^{\Lambda M} = \frac{1}{3}\ln\left(\frac{\Omega_{M0}}{\Omega_{\Lambda 0}}\right).$$

For the derivations of the three x's, see Appendix B.

1.2. Implementation details

1.2.1. Fiducial cosmological parameters

Our model Universe will assume ΛCDM fiducial cosmology and be initialised with the best-fit cosmological parameters from the Planck 2018 data (Aghanim et al. 2020). We will thus take the dimensionless Hubble parameter $h = 0.67$, the current day temperature of the CMB radiation $T_{\rm CMB0} = 2.7255$ K, and the effective number of massless neutrino species $N_{\rm eff} = 3.046$.

For the current day values of the density parameters, we set

$$\begin{aligned}
\Omega_{b0} &= 0.05,\\
\Omega_{\rm CDM0} &= 0.267,\\
\Omega_{K0} &= 0,\\
\Omega_{\gamma 0} &= \frac{\pi^2}{15}\frac{(k_b T_{\rm CMB0})^4}{\hbar^3 c^5}\frac{8\pi G}{3H_0^2} \simeq 5.509\times 10^{-5},\\
\Omega_{\nu 0} &= \frac{7}{8}\left(\frac{4}{11}\right)^{4/3} N_{\rm eff}\Omega_{\gamma 0} \simeq 3.811\times 10^{-5},\\
\Omega_{\Lambda 0} &= 1 - \sum_{i\neq\Lambda}\Omega_{i0} \simeq 0.683.
\end{aligned}$$

Here, the dark energy density parameter is defined such that $\sum_i \Omega_{i0} = 1$, to ensure a spatially flat Universe.

1.2.2. Numerical integration scheme

To evaluate integral quantities such as Eqs.(6) and (7), we recast them as ordinary differential equations and integrate numerically using the fourth-order Runge-Kutta solver provided by the GSL library (GNU Project 2024) and spline the result using the cubic spline algorithm available in GSL.

The cosmic time ODE is

$$\frac{{\rm d}t}{{\rm d}x} = \frac{1}{H(x)},$$

which is to be integrated up from $x_i = -\infty$ to some final x_f. However, in practice, we must choose a very early time to integrate from. In our case, we solve in the interval $x_i = -20$ up to $x_f = 5$. The initial condition for the ODE is determined deep inside the radiation dominated era, when $H(x) \sim H_0\sqrt{\Omega_{R0}}e^{-2x}$, such that

$$\frac{{\rm d}t}{{\rm d}x} = \frac{e^{2x}}{H_0\sqrt{\Omega_{R0}}} \;\Rightarrow\; t(x_i) = \int_{-\infty}^{x_i}\frac{e^{2x}{\rm d}x}{H_0\sqrt{\Omega_{R0}}} = \frac{1}{2H(x_i)}.$$

Similarly, we solve the ODE for the conformal time

$$\frac{{\rm d}\eta}{{\rm d}x} = \frac{c}{\mathcal{H}(x)}$$

in the same interval, with the initial condition $\eta(x_i) = c/\mathcal{H}(x_i)$, due to the radiation dominated conformal Hubble function $\mathcal{H}(x) \sim H_0\sqrt{\Omega_{R0}}e^{-x}$.

1.2.3. MCMC parameter fitting

Using Markov Chain Monte Carlo (MCMC), we will test our simulated Universe, based on fiducial cosmological parameters, against observations of type Ia supernova (SNIa) luminosity distances (Betoule et al. 2014). The procedure searches for the triplet of "best-fit" parameters in the $(h, \Omega_{M0}, \Omega_{K0})$ parameter space that minimises

$$\chi^2(h,\Omega_{M0},\Omega_{K0}) = \sum_{i=1}^{N}\frac{[d_L(z_i,h,\Omega_{M0},\Omega_{K0}) - d_L^{\rm obs}(z_i)]^2}{\sigma_i^2}.$$

Here, d_L are the predicted luminosity distances from our model at the observed redshifts z_i, while $d_L^{\rm obs}$ are the observed distances, together with their associated errors.

1.3. Results

1.3.1. Timeline of the background

In Table 1, we have stated the times for the matter-radiation equality, the onset of cosmic expansion, and the matter-dark energy equality, in terms of the log scale factor x, the redshift $z(x) = e^{-x} - 1$ and the cosmic time $t(x)$. We remark that cosmic acceleration only begins in the very late Universe at around $z \sim 0.6$. We thus observe the same late-stage acceleration as we would expect from the ΛCDM Universe.

Displayed in Figure 2 are the density parameters Ω_R, Ω_M, and Ω_Λ as functions of x, representing the contributions from relativistic components, non-relativistic matter, and the cosmological constant. The figure highlights three distinct periods in the history of our model Universe:

(i) an era of radiation domination in the very early Universe until $x_{\rm eq}^{MR} = -8.132$;

(ii) a matter dominated period between $x_{\rm eq}^{MR} = -8.132$ and $x_{\rm eq}^{\Lambda M} = -0.256$;

(iii) and a late stage dark energy dominated Universe after $x_{\rm eq}^{\Lambda M} = -0.256$.

This timeline is explained by the fact that $\rho_R \propto a^{-4}$, $\rho_M \propto a^{-3}$, while $\rho_\Lambda = $ const, such that the density of the relativistic components decays faster than non-relativistic matter, which in turn decays faster than the density of dark energy.

Equality times and onset of acceleration			
Event	Log scale factor x	Redshift z	Time t [Gyr]
Matter-radiation equality (MR)	-8.132	3400.33	5.106×10^{-5}
Onset of acceleration (acc)	-0.487	0.6272	7.752
Matter-dark energy equality (ΛM)	-0.256	0.2915	10.378

Table 1. Times for when we have matter-radiation equality, onset of cosmic acceleration, and matter-dark energy equality expressed using x, redshift $z = e^{-x} - 1$, and cosmic time $t(x)$.

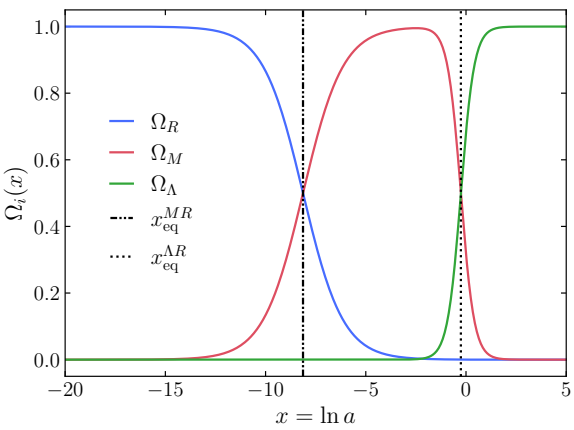

Fig. 2. Evolution of the density parameters of relativistic species (Ω_R), non-relativistic matter (Ω_M), and of dark energy (Ω_Λ) as functions of $x = \ln a$ on the interval $-20 \leq x \leq 5$. Plotted with vertical dashed and dotted lines are also the times of matter-radiation and matter-dark energy equality.

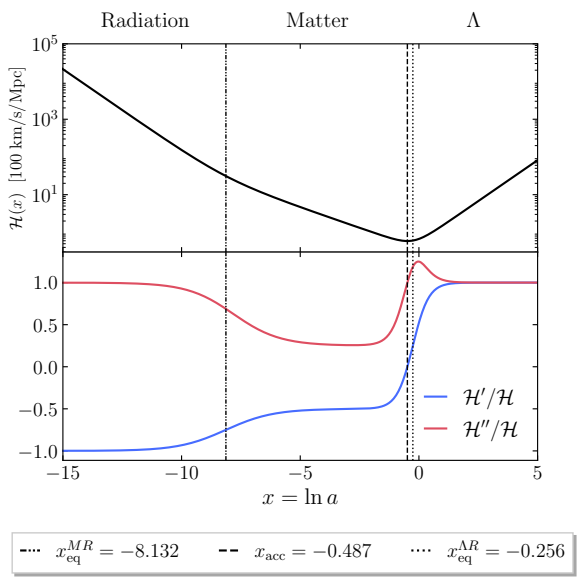

Fig. 3. The conformal Hubble function $\mathcal{H}(x)$ along with $\mathcal{H}'/\mathcal{H}$ and $\mathcal{H}''/\mathcal{H}$ plotted as functions of x. The time at which we have the onset of cosmic acceleration $x_{\rm acc}$, and the equality times $x_{\rm eq}^{mr}$ and $x_{\rm eq}^{\Lambda m}$ are shown with vertical dotted lines with the dominating component indicated above the regions in between them.

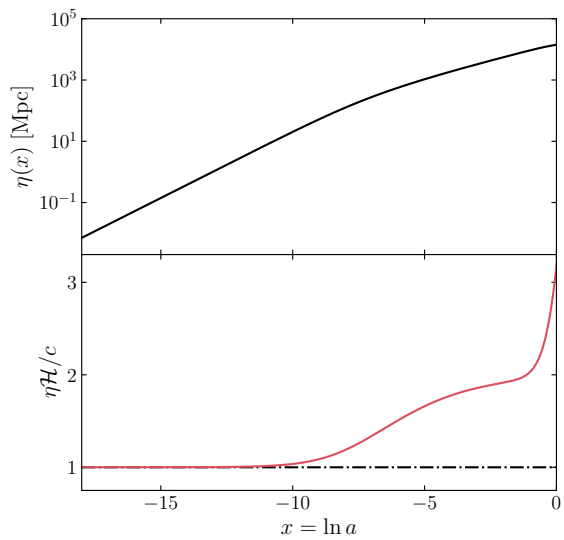

Fig. 4. The top panel presents the evolution of the conformal time with respect to x. The bottom panel shows the dimensionless quantity $\eta\mathcal{H}/c$ over the same interval.

The transition between cosmological eras leaves clear signatures in the conformal Hubble function and its derivatives, as illustrated in Figure 3. Equations (8) and (9) predict specific values in each regime: $\mathcal{H}'/\mathcal{H} = -1$ and $\mathcal{H}''/\mathcal{H} = 1$ in radiation domination, $\mathcal{H}'/\mathcal{H} = -1/2$ and $\mathcal{H}''/\mathcal{H} = 1/4$ in matter domination, and both ratios approaching unity in a Λ-dominated universe. These predicted features are apparent in the bottom panel of Figure 3.

The top panel also exhibits the cosmic acceleration after $x_{\rm acc}$. Since $\mathcal{H} \equiv aH = \dot{a}$, the conformal Hubble function describes the rate of change of the scale factor with respect to cosmic time. We observe that the slope of $\mathcal{H}$ changes from negative to positive, i.e. $\mathcal{H}'/\mathcal{H}$ becomes positive, at the minimum situated at $x = x_{acc}$. A positive slope means that

$$\mathcal{H}' = \frac{\mathrm{d}\dot{a}}{\mathrm{d}x} = \frac{\mathrm{d}t}{\mathrm{d}x}\ddot{a} > 0.$$

Since $t(x)$ is manifestly a strictly increasing function, $\mathrm{d}t/\mathrm{d}x$ is strictly positive at all times, such that the sign of $\mathcal{H}'$ depends on the sign of $\ddot{a}$. Therefore, we see that $\ddot{a} > 0$ for $x > x_{\rm acc}$, so we indeed have late-stage cosmic acceleration.

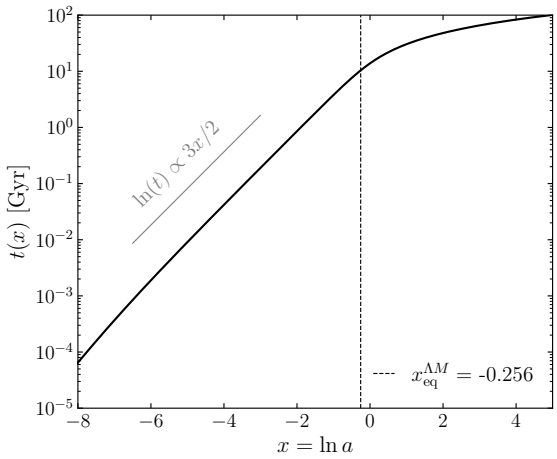

Fig. 5. Cosmic time with respect to x. The gray diagonal line is included in order to compare the slope of $t(x)$ with our expectation of $\ln t \propto 3x/2$ during matter domination.

1.3.2. Cosmic and conformal time, and the age of the Universe

Figure 4 presents the conformal time $\eta(x)$ as well as the dimensionless quantity $\eta\mathcal{H}/c$ in the interval $-18 \leq x \leq 0$. From the initial condition $\eta(x_i) = c/\mathcal{H}(x_i)$, derived in Section 1.2.2, we expect that $\eta\mathcal{H}/c \to 1$ as $x \to -\infty$. We observe that this is in fact the case for our model (see top panel, Fig.4). The conformal time is also a smooth monotonically increasing function of x, and its current day value is calculated to be $\eta(0)/c \simeq 46.319$ Gyr.

The solution for the cosmic time $t(x)$ is showcased in Figure 5 over the interval $-8 \leq x \leq 5$, with a current day value determined to be $t_0 \equiv t(0) \simeq 13.858$ Gyr. During the matter domination era, the Hubble function is $H(x) \propto e^{-3x/2}$, so we would expect that

$$\frac{\mathrm{d}t}{\mathrm{d}x} \propto e^{3x/2} \;\Rightarrow\; \ln t(x) \propto \frac{3}{2}x.$$

Comparing to the gray line included in Figure 5, we see that it matches the slope of $t(x)$ between $x = -8$ up to $x \sim x_{\rm eq}^{\Lambda M}$. In the dark energy dominated era ($x \gtrsim x_{\rm eq}^{\Lambda M}$), the slope decreases and the curve flattens out. This is to be expected from the fact that $H \sim H_0\sqrt{\Omega_{\Lambda 0}} = \text{const}$ such that $t \propto x$ during this era.

1.3.3. MCMC best-fit parameters and the Hubble tension

The MCMC parameter fitting to SNIa luminosity distance data (see Sect.1.2.3) yielded the best-fit triplet of parameters

$$h = 0.701711,$$
$$\Omega_{M0} = 0.255027,$$
$$\Omega_{K0} = 0.0789514,$$

with an associated $\chi^2 = 29.2799$. The observational data contains $N = 31$ data, so we have $\chi^2/N \lesssim 1$, indicating a good fit.

Figure 6 shows the $(\Omega_{M0}, \Omega_{\Lambda 0})$ parameter space sampled during the MCMC procedure, highlighting the regions within 1σ and 2σ of the best-fit parameters (minimum χ^2). The fiducial Planck values lie within the 1σ contour, though slightly offset from the best-fit point.

The marginalized posterior distributions for $\Omega_{\Lambda 0}$ and the Hubble parameter h are presented in Figure 7, with vertical dotted lines indicating the fiducial Planck values. While the fiducial

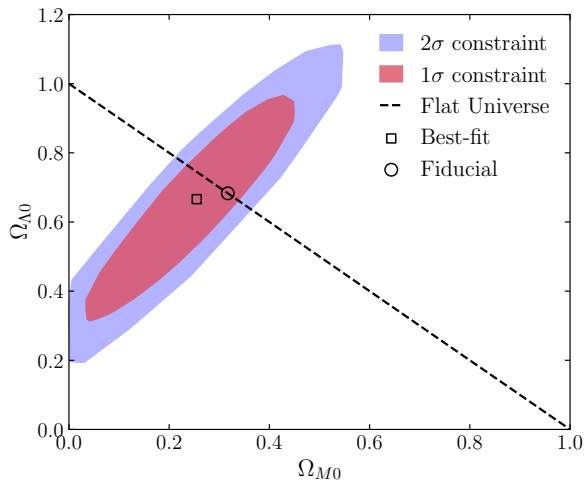

Fig. 6. The regions of sampled points in the $(\Omega_{M0}, \Omega_{\Lambda 0})$ parameter space within 1σ and 2σ of the pair associated with the minimal χ^2. The diagonal dashed line shows the cosmologies giving a flat Universe; $\Omega_{K0} = 1 - \Omega_{M0} - \Omega_{\Lambda 0} = 0$.

$\Omega_{\Lambda 0}$ falls comfortably within the posterior and close to its best-fit value, the same is not true for h. The fiducial value of h lies well outside the 1σ region of its posterior distribution, with a relative difference of

$$\left|\frac{h_{\text{best-fit}} - h_{\text{fiducial}}}{h_{\text{best-fit}}}\right| \simeq 4.5\%.$$

This discrepancy is further illustrated in Figure 8, which compares the luminosity distance-redshift relation predicted by our best-fit model to that of the fiducial Planck cosmology, alongside the observed SNIa data. The best-fit parameters clearly provide a superior match to the supernova measurements across the full redshift range, reinforcing that while the Planck cosmology remains our adopted baseline for subsequent calculations, it is less consistent with these late-universe distance indicators.

Together, these results highlight the well-known *Hubble tension*: the persistent disagreement between the value of H_0 inferred from early-universe observations (such as the CMB) and that derived from late-universe distance measurements (e.g., supernovae or galaxy surveys). Despite this tension, we will adopt the fiducial Planck ΛCDM cosmology for the remainder of this work, while noting its inconsistency with the SNIa luminosity distances used in our MCMC analysis.

2. Recombination history

Some introduction about what it is all about.

2.1. Theory

The theory behind this milestone.

2.2. Implementation details

Something about the numerical work.

2.3. Results

Show and discuss the results.

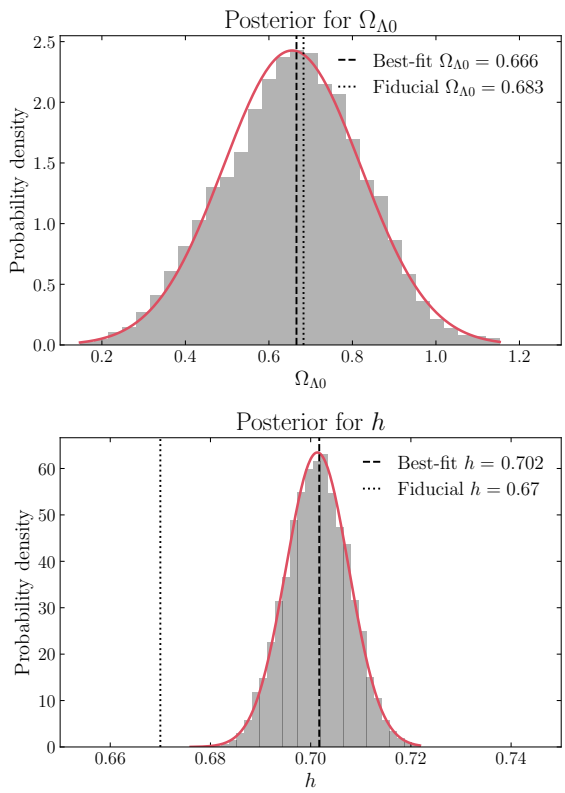

Fig. 7. Posteriors for $\Omega_{\Lambda 0}$ and h from the MCMC parameter fitting. In the posterior for h, we have also indicated the fiducial value of $h = 0.67$ with a blue vertical line to highlight the Hubble tension.

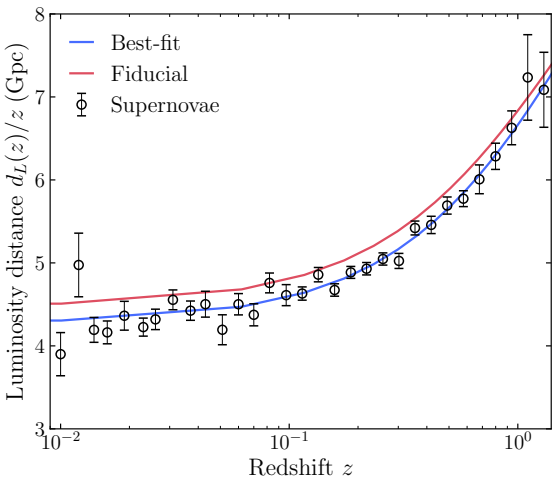

Fig. 8. Luminosity distance as a function of redshift z. The data points represent the luminosity distances deduced from observations of supernovae at different redshifts. The two curves show the luminosity distance functions $d_L(z)/z$ for the best fit parameters and for the fiducial cosmological parameters.

3. Conclusions

I will write a short summary and conclusion in the end.

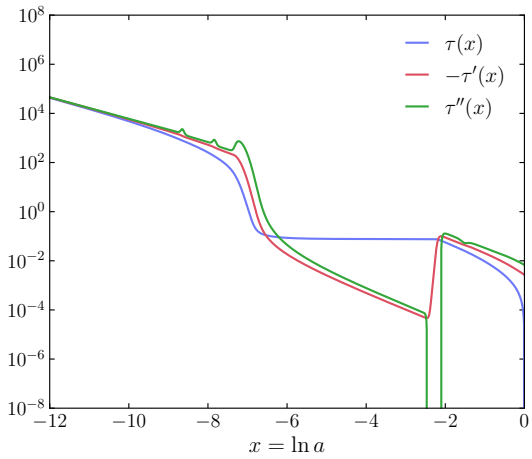

Fig. 9. yep

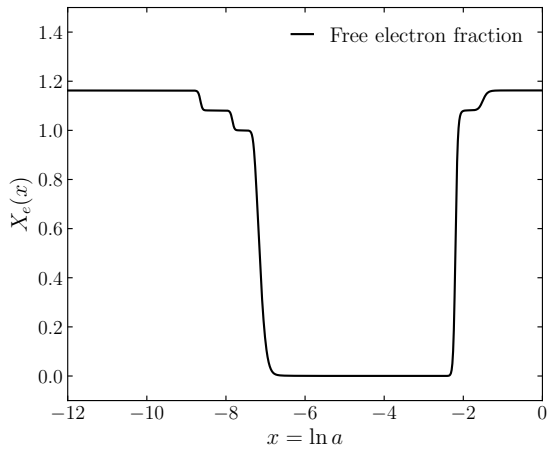

Fig. 10. yep.

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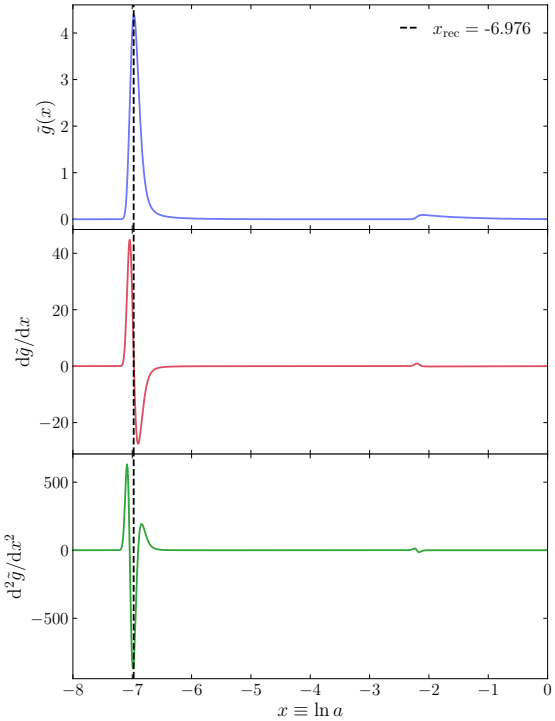

Fig. 11. No

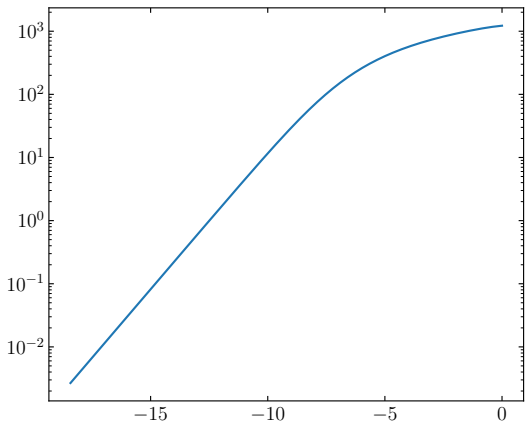

Fig. 12. No

Appendix A: Derivatives of the conformal Hubble function

The Hubble function is given by

$$H = H_0 \sqrt{(\Omega_{b0} + \Omega_{\rm CDM0})a^{-3} + (\Omega_{\gamma 0} + \Omega_{\nu 0})a^{-4} + \Omega_{k0}a^{-2} + \Omega_{\Lambda 0}},$$

where we can define $\Omega_{M0} \equiv \Omega_{b0} + \Omega_{\rm CDM0}$ and $\Omega_{R0} \equiv \Omega_{\gamma 0} + \Omega_{\nu 0}$. The first derivative of the conformal Hubble function $\mathcal{H} \equiv aH(a)$ with respect to $x \equiv \ln a$ is

$$\frac{\mathrm{d}\mathcal{H}}{\mathrm{d}x} = \frac{\mathrm{d}a}{\mathrm{d}x}H + a\frac{\mathrm{d}H}{\mathrm{d}x} = \mathcal{H} + a^2\frac{\mathrm{d}H}{\mathrm{d}a}.$$

Let us compute the derivative of $H(a)$:

$$\frac{\mathrm{d}H}{\mathrm{d}a} = -\frac{H_0^2}{2H}\Big(3\Omega_{M0}a^{-4} + 4\Omega_{R0}a^{-5} + 2\Omega_{k0}a^{-3}\Big).$$

This gives

$$\begin{aligned}\frac{\mathrm{d}\mathcal{H}}{\mathrm{d}x} &= \mathcal{H} - a^2\frac{H_0^2}{2H}\Big(3\Omega_{M0}a^{-4} + 4\Omega_{R0}a^{-5} + 2\Omega_{k0}a^{-3}\Big)\\ &= \mathcal{H}\Big[1 - \frac{3}{2}\Omega_M - 2\Omega_R - \Omega_K\Big].\end{aligned}$$

Here the Ω_i's refer to the density parameters as functions of scale factor, i.e. $\Omega_i(a)$, which should be evaluated at $a = e^x$.

The second derivative is a bit more cumbersome to calculate; from the product rule it follows that

$$\begin{aligned}\frac{\mathrm{d}^2\mathcal{H}}{\mathrm{d}x^2} &= \mathcal{H}\Big[1 - \frac{3}{2}\Omega_m - 2\Omega_r - \Omega_K\Big]^2\\ &\quad + \mathcal{H}\frac{\mathrm{d}}{\mathrm{d}x}\Big[1 - \frac{3}{2}\Omega_m - 2\Omega_r - \Omega_K\Big].\end{aligned}$$

Let B be the expression inside the big bracket. We thus need $\mathrm{d}B/\mathrm{d}x$, which is most easily calculated by factoring out $H_0^2/H(a)^2$ from the last three terms of B;

$$\begin{aligned}\frac{\mathrm{d}B}{\mathrm{d}x} &= -\frac{a}{2}\frac{\mathrm{d}}{\mathrm{d}a}\left(\frac{H_0^2}{H^2}\right)\Big(3\Omega_{M0}a^{-3} + 4\Omega_{R0}a^{-4} + 2\Omega_{K0}a^{-2}\Big)\\ &\quad - \frac{a}{2}\frac{H_0^2}{H^2}\frac{\mathrm{d}}{\mathrm{d}a}\Big(3\Omega_{M0}a^{-3} + 4\Omega_{R0}a^{-4} + 2\Omega_{K0}a^{-2}\Big)\\ &= -\frac{1}{2}\Big(3\Omega_M + 4\Omega_R + 2\Omega_K\Big)^2 + \frac{1}{2}\Big(9\Omega_M + 16\Omega_R + 4\Omega_K\Big).\end{aligned}$$

From this we end up with second derivative

$$\begin{aligned}\frac{\mathrm{d}^2\mathcal{H}}{\mathrm{d}x^2} &= \mathcal{H}\Bigg\{\Big[1 - \frac{3}{2}\Omega_M - 2\Omega_R - \Omega_K\Big]^2\\ &\quad - \frac{1}{2}\Big(3\Omega_M + 4\Omega_R + 2\Omega_K\Big)^2 + \frac{1}{2}\Big(9\Omega_M + 16\Omega_R + 4\Omega_K\Big)\Bigg\}\Bigg|_{a=e^x}.\end{aligned}$$

Appendix B: Logarithmic scale factor at different cosmological events

Matter-radiation equality happens when $\Omega_M(a_{\rm eq}^{MR}) = \Omega_R(a_{\rm eq}^{MR})$. Using the expressions given in Eq.(5), we find

$$\frac{\Omega_{M0}}{(a_{\rm eq}^{MR})^3} = \frac{\Omega_{R0}}{(a_{\rm eq}^{MR})^4} \;\Rightarrow\; x_{\rm eq}^{MR} \equiv \ln a_{\rm eq}^{MR} = \ln\left(\frac{\Omega_{R0}}{\Omega_{M0}}\right).$$

In a similar way, we derive that the matter-dark energy equality occurs for

$$\frac{\Omega_{M0}}{(a_{\rm eq}^{\Lambda M})^3} = \Omega_{\Lambda 0} \;\Rightarrow\; x_{\rm eq}^{\Lambda M} \equiv \ln a_{\rm eq}^{\Lambda M} = \frac{1}{3}\ln\left(\frac{\Omega_{M0}}{\Omega_{\Lambda 0}}\right).$$

The onset of cosmic acceleration is defined to be when $\ddot{a} = 0$. From the second Friedmann equation, Eq.(3), we get that this occurs whenever

$$\sum_i \rho_i(a_{\rm acc})(1 + 3w_i) = 0.$$

The relevant equations of state are $w_M = 0$, $w_R = 1/3$ and $w_\Lambda = -1$. We neglect the curvature parameter when working with fiducial cosmological parameter, and thus the condition for cosmic acceleration can be recast into

$$\begin{aligned}&\Omega_M + 2\Omega_R - 2\Omega_\Lambda = 0\\ \Rightarrow\quad &\frac{\Omega_{M0}}{(a_{\rm acc})^3} + \frac{2\Omega_{R0}}{(a_{\rm acc})^4} - 2\Omega_{\Lambda 0} = 0.\end{aligned}$$

At late times, the radiation density parameter can be neglected, so we end up with

$$x_{\rm acc} \equiv \ln a_{\rm acc} = \frac{1}{3}\ln\left(\frac{\Omega_{M0}}{2\Omega_{\Lambda 0}}\right).$$